

GA4 — How Analytics Works on This Site

Site: <https://www.neuro-stellar.com> — Next.js 16 (App Router), React 19, Vercel. **GA4 property:** fresh property `G-F75KF0N3SH` (not the old Wix `G-KEVS5H2JQN`). **Status:** built and in the code. The only thing left is operational — set the env var and finish the GA4 Admin config (Sections 5–6). **Companion doc:** `utm-gsc-clarity-plan.md`.

1. How the tag loads

- `@next/third-parties/google` renders the GA tag in `app/layout.tsx`:

```
{GA_ID} && <Analytics gaId={GA_ID} /> // GA_ID = process.env.NEXT_PUBLIC_GA_ID
```

- It loads **only when** `NEXT_PUBLIC_GA_ID` is set — locally in `.env.local`, and in Vercel (Production + Preview), then redeploy.
- Vercel Analytics (`<Analytics />`) runs alongside it. Every custom event fires to **both** sinks.
- App Router page views are handled automatically by the tag + Enhanced Measurement. We do **not** hand-roll a pageview listener.

2. Consent Mode v2

GA sets no cookies before consent.

- A `beforeInteractive` script in `app/layout.tsx` sets all four signals to `denied` by default (`ad_storage`, `analytics_storage`, `ad_user_data`, `ad_personalization`).
- The cookie banner (`components/CookieConsent.tsx` + `lib/consent.ts`) flips them to `granted` on Accept and stores the choice in `localStorage`.
- Re-open the banner anytime via the **Cookie settings** link in the footer.
- File the GA tag under the **analytics** consent category in GA4.

3. Firing events — `lib/analytics.ts`

One typed wrapper dual-fires to GA4 + Vercel:

```

trackEvent(name, params)           // base
trackCTA(action, location)         // → cta_click
trackScheduleDemo()               // → schedule_demo
trackCareersApply(jobId, title)    // → careers_apply
trackFormError(formId, message)    // → form_error
trackContactClick(method, loc)     // → contact_click
trackFaqOpen(question)            // → faq_open
trackReportTourChapter(ch, i)     // → report_tour_chapter

```

4. Events we send

snake_case. Google recommended names where one fits, custom otherwise.

Event	Params	Key event	Where it fires
page_view	page_location, page_title	—	Enhanced Measurement (auto)
cta_click	cta_action, cta_location	—	BookDemoButton.tsx
schedule_demo_start	method	—	Calendly modal opens (CalendlyModal.tsx)
schedule_demo	method	✓	Calendly booking confirmed
careers_apply	job_id, job_title	✓	Careers form success (CareersForm.tsx)
form_error	form_id, message	—	Careers form failure
contact_click	method, contact_location	—	any tel: / mailto: click (ContactLinkTracker.tsx)
faq_open	question	—	FAQ accordion (Accordion.tsx trackAs="faq")
report_tour_chapter	chapter, chapter_index	—	/report chapter scrolls into view
scroll	percent_scrolled	—	Enhanced Measurement (auto)
click (outbound)	link_url, link_domain	—	Enhanced Measurement (auto) — covers social/news/partner links

Conversion model: demo-first, no checkout. The conversion is a **booked Calendly meeting** (schedule_demo), opened by any "Book a Demo" button. Careers applications are the second key event.

Not tracked on purpose: self-hosted video, mobile-menu toggles, nav clicks, and the Spline iframe — low signal or not visible to GA4. Outbound clicks are left to Enhanced Measurement.

5. Custom dimensions (Admin → Custom definitions)

Event-scoped, snake_case, never PII. Register the params you want to report on:

`cta_action`, `cta_location`, `method`, `contact_location`, `job_id`, `job_title`, `question`, `chapter`.

New dimensions read "(not set)" for ~24–48 h before populating.

6. GA4 Admin checklist

Setting	Action
Data stream	Web stream for <code>neuro-stellar.com</code> under property G-F75KF0N3SH
Enhanced Measurement	ON (keep history-events, outbound clicks, scroll, file downloads)
Key events	Mark <code>schedule_demo</code> and <code>careers_apply</code>
Custom dimensions	Register the 8 params above
Data retention	2 → 14 months
Internal traffic	Filter the team IP(s), set Active
Google signals	OFF at this traffic
Consent category	GA tag under analytics
Link Search Console	Admin → Product links (see companion doc)

7. QA

On the deployed site (not localhost), in GA4 **DebugView / Realtime**:

1. Accept the banner → `page_view` fires once per navigation.
2. Click a CTA → `cta_click` with the right `cta_action` / `cta_location`.
3. Book a test Calendly slot → `schedule_demo` (and `schedule_demo_start` when the modal opens).
4. Submit the careers form → `careers_apply`; open an FAQ → `faq_open`; scroll /report → `report_tour_chapter`; click a phone/email link → `contact_click`.
5. Decline consent in a fresh session → no analytics cookies, no hits.

Appendix — GA4 quick reference

- **Event tiers:** automatic · enhanced measurement · recommended (Google's names) · custom. Prefer a recommended name when one fits.

- **Key events** = the 2024 rename of "conversions". Build Google Ads conversions from the GA4 key event later if you run paid.
- **Custom dimensions:** register a parameter to report on it. 50 event / 25 user scoped. "(not set)" for ~48 h after creation.
- **Consent Mode v2 signals:** `ad_storage`, `analytics_storage`, `ad_user_data`, `ad_personalization`. Required since Mar 2024 for EEA/UK Google tags.
- **Low traffic:** under ~1,000 events/day, behavioral modeling won't qualify — get the events right and lean on Clarity for qualitative insight (companion doc).